5. Each polynucleotide chain is made up of four different bases. The purine bases present in DNA are adenine and guanine and the pyrimidine bases present are thymine and cytosine. The sequence of purine and pyrimidine carry the genetic information whereas the sugar and phosphate groups perform the structural role.

6. Each polynucleotide chain has direction or polarity. Further, each polynucleotide chain has 5′phosphorylated and 3′hydroxyl ends.

7. The two strands run in opposite direction (i.e.) they are antiparallel.

8. The two strands are held together by hydrogen bonds (base pairing) between the purine and pyrimidine bases of the opposite strands.

Watson and Crick deduced the rules of base pairing (Fig. 7.7). They are:

- The purine adenine (A) always pairs with the pyrimidine thymine (T).
- The purine guanine (G) always pairs with the pyrimidine cytosine (C).

Base pairing is achieved through hydrogen bonding.

Therefore, if adenine appears in one strand, thymine is found in the opposite strand and vice versa. When guanine is found in one strand, cytosine is present in the opposite strand and vice versa. So, Base sequence of one strand is complementary to the opposite strand. For example, If the sequence 5'ATGGACC3′is present in one strand, the complementary strand will be having the sequence, 3'TACCTGG5'.

There are two hydrogen bonds between A and T and three hydrogen bonds between G and C.

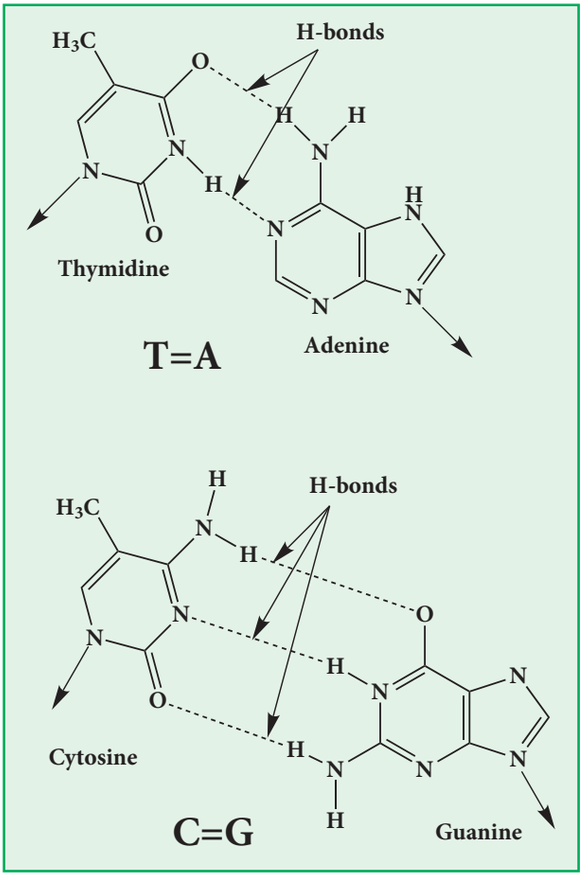

Figure. 7.7 Nucleotide base pairing

9. Base composition of DNA obeys Chargaff's rule.

Chargaff's rule

Erwin Chargaff, a scientist

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